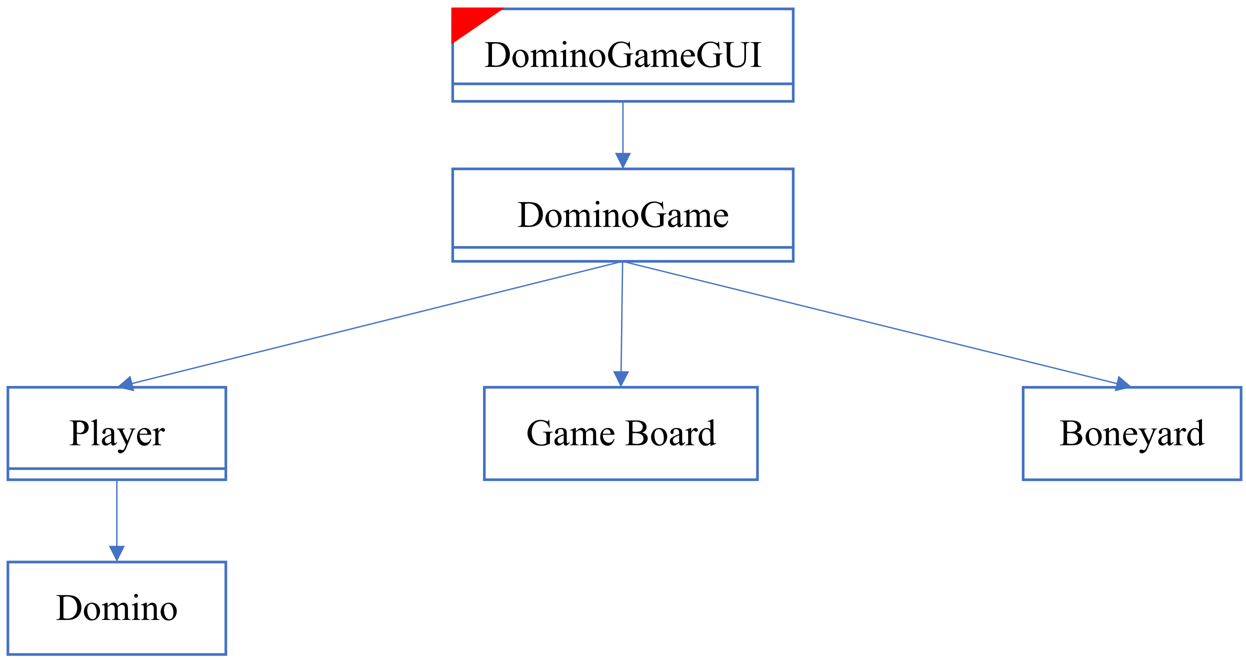
Proposed Design of Version 2



Trigger: update from mouse and keyboard

Description of the Design

DominoGameGUI : Manages the graphical user interface for the Domino game.

- Data:
  - game: The main game logic object
  - boardPanel: Panel for displaying the game board
  - playerHandPanel: Panel for displaying the player's hand
  - drawButton: Button for drawing dominos
  - messageLabel: Label for displaying game messages
- Behavior:
  - initializeUI(): Initialize the user interface components
  - drawBoard(Graphics): Draws the game board with the current state of played dominos.
  - drawDomino(Graphics, Domino, int, int): Draw a single domino on the board
  - updateUI(): Updates all UI components to reflect the current game state
  - updatePlayerHand(): Updates the display of the player's hand
  - playDomino(Domino): Handles the logic when a player attempts to play a domino
  - updateMessage(messageLabel): Updates the message displayed to the user.

DominoGame : Implements the core game logic and manages the game flow.

- Data:
  - human: Represents the human player
  - computer: Represents the computer player
  - boneyard: The pool of undrawn dominos
  - board: the current state of the game board
  - lastPlayer: Player who made the last move
  - gameOver: Indicates if the game has ended
- Behavior:
  - initializeGame(): Initializes a new game.
  - dealInitialHands(): Distributes initial dominos to players.
  - drawForHuman(): Allows the player to take a domino from the boneyard.
  - playHumanDomino(Domino, boolean, boolean): Process a human player's move
  - computerTurn(): Manages the computer's turn
  - canPlay(Domino, boolean): Checks if a domino can be played
  - isValidPlay(Domino, boolean): Checks if a play is valid
  - checkGameOver(): Checks if the game is over
  - hasPlayableDomino(Player): Checks if a player has a playable dominos.
  - calculateFinalScore(Player): Calculates the final score for a player.
  - getBoardDominos(): Gets the list of dominos currently on the board.
  - getHumanHand(): Gets the human player’s hand
  - canHumanDraw(): Checks if the human player can draw a domino
  - isGameOver(): Checks if the game is over
  - getGameStateMessage(): Gets the current game state message
  - resetGame(): Resets the game to initialize the game

GameBoard: Represents the playing area where dominos are placed.

- Data:
  - dominos: List of Domino objects on the board.
  - dominoPositions: List of positions for each played domino.
  - leftEnd: int representing the value on the left end of the board.
  - rightEnd: int representing the value on the right end of the board.
- Behavior:
  - GameBoard(): Initializes an empty game board

- playDomino(Domino, boolean): Plays a domino to the board.
- getLeftEnd(): Gets the value on the left end of the board.
- getRightEnd(): Gets the value on the right end of the board.
- getDominos(): Gets a copy of the list of dominos on the board.
- getDominoPositions(): Returns the list of domino positions.
- isEmpty(): Checks if the board is empty.

Player: Represents a player in the game.

- Data:
  - hand: The dominos in the player´s hand
  - name: String representing the player´s name.
- Behavior:
  - Player(String): Initializes a player with a given name
  - addDomino(Domino): Adds a domino to the player's hand
  - removeDomino(Domino): Removes a domino from the player´s hand.
  - getHand(): Gets a copy of the player's hand.

Boneyard: Represents the pool of undrawn dominos.

- Data:
  - dominos: List of dominos in the boneyard
- Behavior:
  - Boneyard(): Initializes the boneyard with a full set of dominos and shuffles them.
  - draw(): Draws a domino from the boneyard.
  - isEmpty(): Checks if the boneyard is empty.

Domino: Represents a single domino piece.

- Data:
  - left: int representing the value on the left side.
  - right: int representing the value on the right side.
- Behavior:
  - Domino(int, int): Creates a new domino with given left and right values.
  - getLeft(): Gets the left value of the domino.
  - getRight(): Gets the right value of the domino.
  - rotate(): Rotates the domino by swapping its left and right values.
  - getSum(): Calculates the sum of the domino´s values.
  - toString(): Provides a string representation of the domino.